

# IoT-Based Smart Faculty and Classroom Monitoring System Using Rfid Technology and LoRaWAN

Submitted by:

Arguelles, Ron

Ledesma, Aily Kathleen S.

Morante, Harold

Reyes, Jamby

Villasor, Jerome

## Project Overview

### Project Description

The IoT-Based Smart Faculty and Classroom Monitoring System Using RFID Technology and LoRaWAN is designed to monitor the real-time location of faculty members within the school campus. The system uses RFID ID cards that faculty members tap when leaving or entering a classroom. It is designed to track teachers' location and attendance, focusing on whether they are inside a specific classroom or within the faculty office area.

Once the RFID ID is tapped, the information is sent through LoRaWAN communication, allowing the system to update the teacher's current location even across long distances. The updated location is then displayed on a faculty monitoring screen, showing which classroom the teacher is currently in.

This system is not an attendance system. It is designed to improve classroom coordination and faculty management efficiently. It enables the administration to see a "real-time dashboard" to verify if a teacher is currently in the classroom, in the faculty room, or outside the campus.

This system is not an attendance tool, but purely for monitoring and location tracking. The IoT-Based Smart Faculty and Classroom Monitoring System Using RFID Technology and LoRaWAN focuses on monitoring the real-time location of faculty members inside the school campus. The system uses RFID technology for identification and LoRaWAN for wireless data communication.

RFID readers are set up at the doors of classrooms and other important places on campus to pick up on when teachers come and go. LoRaWAN sends the information to a central server wirelessly whenever a teacher walks by. The server then updates a live monitoring dashboard that staff members who have permission can check from a computer or mobile device at any time. This makes it easy for them to see who is in which room at any time.

In addition, LoRaWAN (Long Range Wide Area Network) is often used to send this RFID data from many different rooms over a long distance to a central server without needing heavy WiFi infrastructure, which is ideal for large school campuses. This system ensures that the school knows

the exact location of the teacher within the building and ensures they are in their designated room during scheduled hours.

Several studies have looked into how IoT-based monitoring systems that use RFID and LoRaWAN can improve learning spaces. One important study created an RFID-based attendance system for classrooms that worked with the Internet of Things (IoT). RFID tags on student IDs automatically log when they enter and leave, which cuts down on proxy attendance problems and lets the database be updated in real time through web portals. This method is like how faculty and classroom monitoring works by automating tracking and adding sensors for occupancy. It's similar to how LoRaWAN expands connectivity for low-power, long-range data transmission in distributed campus setups.

Researchers have created IoT systems for monitoring faculty and classrooms that go beyond attendance. These systems use RFID tags worn by professors and readers at the entrances to classrooms to track their real-time location and availability. RFID readers use NodeMCU boards to read tag data and update a web dashboard with the current locations of professors on campus. This makes it easier for students to find professors who are free without having to go to their rooms. This setup lets you track things in a way that doesn't invade people's privacy and also makes it easier to use resources, like setting up meetings based on who has confirmed they will be there.

Recent progress in the Internet of Things (IoT) has greatly changed how campuses are run by combining long-range communication and automated identification technologies. Several studies have used Radio Frequency Identification (RFID) to automate the tracking of people in large organizations, which is similar to the system that was suggested. Research in this area shows that traditional Wi-Fi or Bluetooth-based systems often have problems with high power use and short range. However, adding LoRaWAN (Long Range Wide Area Network) makes campus-wide connectivity much better. LoRaWAN can send data over long distances with very little energy use. This means that academic institutions can keep a reliable monitoring system that works well even in places with a lot of buildings.

Moreover, modern studies highlight the transition from basic attendance tracking to advanced real-time location systems (RTLS) to enhance organizational coordination. Research on faculty monitoring reveals that the principal difficulty in school administration lies not solely in documenting attendance, but in addressing the "coordination gap" that arises when staff transition between various classrooms. It has been shown that systems that use microcontrollers like the ESP32 to process RFID data and update central dashboards in real time cut down on "time-waste" by getting rid of the need for manual searching. The Smart Faculty ideas, which focus on real-time location updates, make sure that only authorized people can make smart decisions and make classroom management more efficient overall.

## Problem Statement

In many schools, tracking the real-time location of faculty members during class hours is difficult. When teachers are not in their assigned classrooms, it often causes confusion and delays. Existing systems focus only on attendance and do not provide real-time monitoring, leading to inefficient coordination among staff.

The study aims to address the following specific problems:

1. Difficulty in locating faculty members during class hours.
2. Delays in classroom activities due to absent or late teachers.
3. Lack of real-time monitoring system within the campus.
4. Inefficient communication and coordination among school staff.
5. Time wasted in manually searching for teachers.

## Objectives of the Study

### General Objectives

To develop a system that allows faculty members to tap their RFID ID when entering or leaving classrooms.

## Specific Objectives

The study aims to achieve the following specific Objectives:

1. To track and update the current location of faculty members across different classrooms.
2. To transmit data wirelessly using LoRaWAN for real-time monitoring.
3. To provide a monitoring screen where authorized personnel can easily view faculty locations.
4. To improve classroom coordination and faculty management efficiently.
5. To design a simple, cost-effective, and reliable system that can function without being an attendance tracking tool.

The main goal of this study is to design and implement an IoT-Based Smart Faculty and Classroom Monitoring System Using RFID Technology and LoRaWAN that can help monitor faculty locations in real-time.

## SCOPE AND LIMITATION

### Scope of the Study

This study focuses on the design and implementation of a IoT-Based Smart faculty and classroom monitoring system using rfid technology and LoRaWAN for the IT faculty and student.

The scope of the study includes the following:

- Monitor faculty members' current location inside the school campus.
- Use RFID ID cards for faculty identification.
- Use LoRaWAN for wireless communication.
- Display faculty location on a monitoring dashboard.
- Focus on monitoring and location tracking, not attendance.

The study is limited to monitoring, access control within the IT Building only

### Limitation of the Study

Despite its intended functionality, the proposed system has several limitations that should be acknowledge:

- Faculty must tap their RFID ID to update location.
- Only works in rooms with installed RFID readers.
- Does not track movement inside a room.

### System Architecture

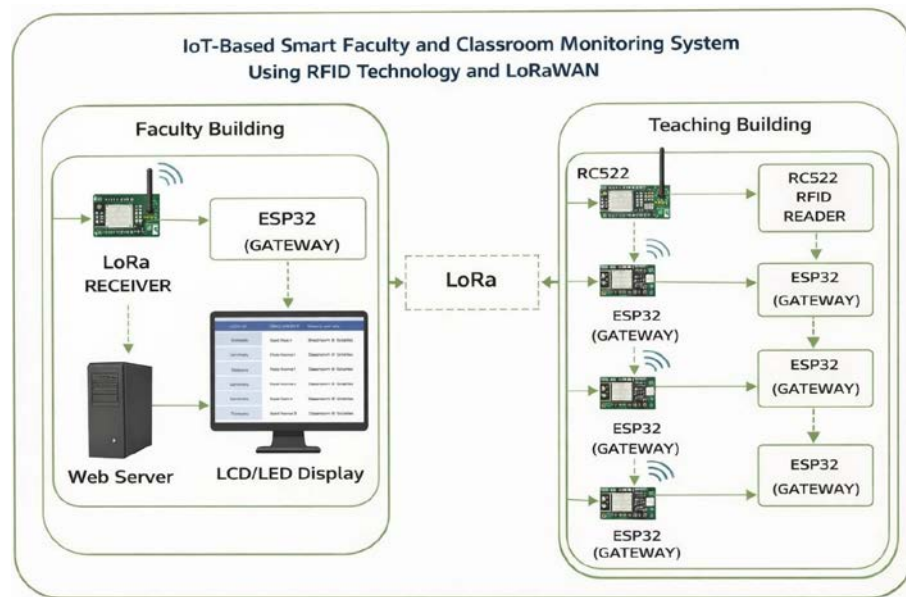


Figure 1: System Architecture

### Architecture Description

The system architecture illustrates the integration of RFID technology, ESP32 microcontrollers, and LoRaWAN communication to monitor the real-time location of faculty members

across the campus. The design follows a distributed IoT architecture composed of two main areas: the Teaching Building and the Faculty Building.

#### Faculty Building (Processing & Output Side)

- 1. LoRa Receiver Module**  
Receives transmitted data from the Teaching Building.
- 2. ESP32 (Gateway Processor)**  
Processes incoming data and forwards it to the server.
- 3. Web Server**  
Stores, manages, and processes faculty location data.
- 4. LCD/LED Monitoring Display**  
Shows real-time faculty location and classroom assignments.

#### Communication Layer

- 5. LoRa Wireless Communication**  
Transmits data from the Teaching Building to the Faculty Building over long distances.

#### Teaching Building (Input Side)

- 6. RC522 RFID Reader**  
Captures the RFID UID when tapped by faculty.
- 7. ESP32 (Gateway Node)**  
Processes the RFID data and prepares it for transmission.
- 8. Multiple RFID Nodes (4 Classrooms)**  
Represents multiple classrooms equipped with RFID readers and ESP32 modules.

## Data Flow

The data flow of the proposed system follows a structured sequence composed of sensing, processing, transmission, and presentation. This ensures accurate, real-time monitoring of faculty location and classroom activity.

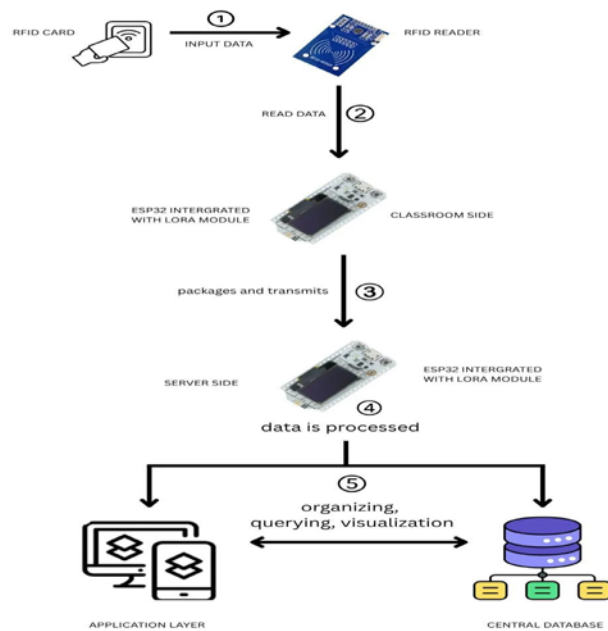


Figure 2: Data Flow

The data flow of the system follows a clear and linear path: **Sensing** → **Data Processing** → **Transmission** → **Reception and Processing (Faculty Building)**. This structured flow ensures data accuracy, reliability, and real-time accessibility while of IoT System Architecture as follows:

### 1. Data Input (Sensing Stage)

The process begins when a faculty member taps an RFID card on the RC522 RFID reader located in the Teaching Building. The reader captures the unique identifier (UID) of the RFID card and sends the data to the ESP32 microcontroller.



## 2. Data Processing in ESP32

The ESP32 processes the captured RFID data by validating and formatting it into a structured data packet. This packet includes relevant information such as user ID, timestamp, and classroom location.

## 3. Data Transmission via LoRa

After processing, the ESP32 transmits the data wirelessly using LoRa communication. This allows long-range and low-power data transfer from the Teaching Building to the Faculty Building.

## 4. Data Reception and Processing (Faculty Building)

The LoRa receiver in the Faculty Building receives the transmitted data. The connected ESP32 (gateway) then decodes and processes the received information before forwarding it to the web server.

## Hardware Components

The system is composed of multiple interconnected hardware components deployed across classrooms and a central faculty monitoring unit. These components work together to enable real-time tracking of faculty location.

| Component       | Description                                                    | Minimum Requirements | Recommended Requirements         |
|-----------------|----------------------------------------------------------------|----------------------|----------------------------------|
| Microcontroller | Central hub for logic, RFID processing, and LoRa communication | ESP8266              | ESP32 DevKit V1                  |
| RFID Reader     | Reads RFID cards and captures unique ID                        | RC522 RFID Module    | RC522 RFID Module (with antenna) |

|                       |                                               |                                |                                                                                           |
|-----------------------|-----------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------|
| RFID Cards/Tags       | Used by faculty for identification            | Standard RFID Cards            | PVC RFID Cards (Durable)                                                                  |
| LoRa Module           | Enables longrange wireless communication      | SX1278                         | SX1278 LoRa Module (High Gain Antenna)                                                    |
| Power Supply          | Provides regulated power to the system        | 5V 2A Adapter                  | 12V 5A Power Supply with Stepdown Converter                                               |
| Wiring & Connectors   | Connects all modules securely                 | Generic Wires                  | Dupont Wires & Soldered Pins                                                              |
| Local Display         | Displays basic system info                    | 16x2 LCD                       | 16x2 LCD with I2C Module                                                                  |
| Chassis Material      | Enclosure for system components               | Plastic/Wood                   | Acrylic or Galvanized Case                                                                |
| Electronics Enclosure | Protects components from damage               | Generic Plastic Box            | ABS Plastic Project Box (IP Rated)                                                        |
| LoRa Antenna          | Improves signal transmission range            | Standard Antenna               | High-Gain LoRa Antenna                                                                    |
| Battery 18650         | Standardized lithium-ion rechargeable battery | Nominal Voltage: 3.6V to 3.7V. | A rating of at least 300–500 full charge/discharge cycles before capacity drops below 80% |

Table 1: Hardware Components

#### Software Components

| Software | Purpose | Minimum Requirements | Recommended Requirements |
|----------|---------|----------------------|--------------------------|
|----------|---------|----------------------|--------------------------|

|                           |                                             |                  |                         |
|---------------------------|---------------------------------------------|------------------|-------------------------|
| Arduino IDE               | Used for programming ESP32 and RFID modules | Basic IDE        |                         |
| Web Server                | Stores and displays faculty location data   | Localhost Server | Cloud-based Server      |
| Database (MySQL/Firebase) | Stores faculty information and logs         | MySQL            | Firebase (Real-time DB) |

Table 2: Software Components

## Functional Requirements

The IoT-Based Smart Faculty and Classroom Monitoring System using RFID Technology and LoRaWAN shall provide the following functionalities:

### 1. RFID Identification

The system shall allow faculty members to identify themselves by tapping their RFID ID cards on the RFID reader installed in each classroom.

### 2. RFID Data Capture

The system shall capture and read the unique identifier (UID) of each RFID card accurately upon tapping.

### 3. Data Processing

The system shall process the captured RFID data using the ESP32 microcontroller and associate it with the corresponding faculty member and classroom.

### 4. Real-Time Location Update

The system shall update the current location of the faculty member immediately after a successful RFID tap.

### 5. LoRaWAN Data Transmission

The system shall transmit the processed data wirelessly from the Teaching Building to the Faculty Building using LoRaWAN technology.

### 6. Data Reception

The system shall receive transmitted data through a LoRa receiver module connected to the ESP32 gateway.

## 7. Data Storage

The system shall store faculty location data, including timestamp and classroom information, in a database.

## 8. Monitoring Dashboard

The system shall provide a monitoring interface where authorized personnel can view the real-time location of faculty members.

## 9. User Access Control

The system shall restrict access to the monitoring dashboard to authorized users only.

## 10. Multiple Classroom Support

The system shall support multiple RFID nodes installed in different classrooms simultaneously.

## 11. System Notification

The system may provide alerts or indicators when a faculty member is not detected in the assigned classroom.

## 12. Data Logging

The system shall log all RFID tap events for record and monitoring purposes.

## Non-Functional Requirements

The IoT-Based Smart Faculty and Classroom Monitoring System using RFID Technology and LoRaWAN shall meet the following non-functional requirements:

### 1. Performance

The system shall update and display faculty location within 2–5 seconds after an RFID tap.

### 2. Reliability

The system shall consistently transmit and receive data accurately with minimal data loss during operation.

### 3. Availability

The system shall be operational during school hours with minimal downtime.

4. Scalability

The system shall support the addition of more classrooms and RFID nodes without significant performance degradation.

5. Security

The system shall ensure that only authorized personnel can access the monitoring dashboard and data.

6. Maintainability

The system shall be easy to maintain, troubleshoot, and update when necessary.

7. Efficiency

The system shall use low power consumption components such as ESP32 and LoRa modules.

8. Data Integrity

The system shall ensure that all recorded data (faculty ID, location, timestamp) is accurate and not altered.

9. Usability

The system shall provide a simple and user-friendly interface for monitoring faculty locations.

10. Compatibility

The system shall be compatible with standard web browsers and commonly used devices.

11. Response Time

The system shall respond to user requests (e.g., viewing dashboard) within a few seconds.

12. Network Coverage

The system shall maintain stable communication across the campus using LoRaWAN technology.

Blynk Virtual Pin Assignment

| Virtual Pin | Datastream<br>Name | Data | Description |
|-------------|--------------------|------|-------------|
|             |                    |      |             |

|    |               |        |                                                                  |
|----|---------------|--------|------------------------------------------------------------------|
| V0 | Faculty ID    | String | Contains the RFID UID of the faculty member who tapped the card. |
| V1 | Faculty Name  | String | Displays the name of the faculty based on the RFID UID.          |
| V2 | Room Location | String | Indicates the classroom                                          |

|    |               |               |                                              |
|----|---------------|---------------|----------------------------------------------|
|    |               |               | where the RFID tap occurred.                 |
| V3 | Timestamp     | String        | Shows the time when the RFID card was tapped |
| V4 | Tap Status    | Integer (0/1) | 1 = Successful tap, 0 = No tap detected.     |
| V5 | System Status | Integer (0/1) | 1 = System ON, 0 = System OFF.               |
| V6 | LoRa Status   | Integer (0/1) | 1 = Connected, 0 = No signal.                |

|    |                    |        |                                                             |
|----|--------------------|--------|-------------------------------------------------------------|
| V7 | Last Detected Room | String | Displays the last recorded classroom of the faculty member. |
|----|--------------------|--------|-------------------------------------------------------------|

## System Operation Procedure

This section describes the step-by-step process of how the IoT-Based Smart Faculty and Classroom Monitoring System operates, from initialization to real-time monitoring.

### Step 1: System Initialization

#### SystemInitialization

The system is powered on, activating all hardware components including the ESP32 microcontroller, RFID reader (RC522), and LoRa module. Each classroom node initializes its devices and prepares for RFID detection. The system ensures that all components are functioning properly before operation begins.

### Step 2: Establish LoRa Communication

#### EstablishLoRaCommunication

After initialization, the ESP32 establishes communication with the LoRa module. The system ensures that all classroom nodes (transmitters) are properly connected to the LoRa gateway

(receiver) located in the Faculty Building. A successful connection allows long-range wireless data transmission.

### Step 3: RFID Tap Detection

#### RFID Tap Detection

When a faculty member enters or exits a classroom, they tap their RFID ID card on the RC522 RFID reader. The reader captures the unique identifier (UID) of the RFID card and sends it to the ESP32 for processing.

### Step 4: Data Processing in ESP32

#### Data Processing in ESP32

The ESP32 processes the captured RFID data by matching the UID with the registered faculty database. It then attaches additional information such as classroom location and timestamp to form a complete data packet.

### Step 5: Data Transmission via LoRa

#### Data Transmission via LoRa

The processed data is transmitted wirelessly from the classroom node (Teaching Building) to the LoRa receiver (Faculty Building). LoRa technology enables long-range and low-power communication across the campus.



## Step 6: Data Reception and Server Update

DataReceptionand ServerUpdate

The LoRa receiver connected to the ESP32 gateway receives the transmitted data. The gateway processes and forwards the data to the web server or monitoring system, where it is stored and updated in real-time.

## Step 7: Monitoring Display

MonitoringDisplay

can view

Authorizedpersonnel

the real-time location of faculty members through the monitoring dashboard or display screen. The system shows the current classroom and last detected location of each faculty member.

## Step 8: Continuous Monitoring

ContinuousMonitoring

The system continuously operates by waiting for new RFID taps. Each tap updates the faculty member's location, ensuring accurate and real-time monitoring throughout the school hours.

# LoRa Monitoring System

## Device Operating Modes

While the system follows a general operational flow, it can be toggled into three specific operating modes to cater to different administrative needs.

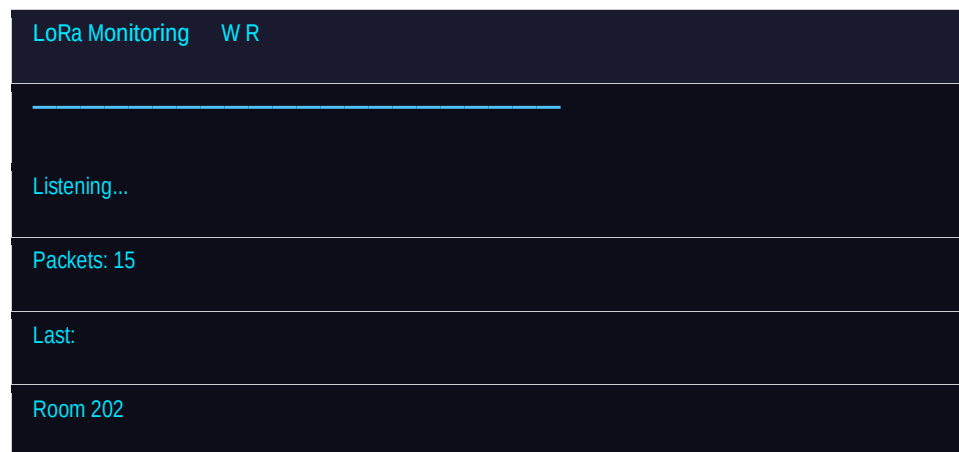
### Mode 1: LoRa Monitoring (Default)

Purpose: Receive attendance data from room transmitters.

#### How It Works

- Listens for LoRa packets from all transmitters
- Displays received packets on OLED
- Uploads data to Firebase automatically
- Shows packet count and last received room

#### OLED Display



## What Happens

1. Transmitter sends: RFID:A3:4F:2E:91:7655:IN:room\_202
2. Receiver gets packet via LoRa
3. Queues to Firebase
4. Dashboard updates in real-time

## Status Indicators

- W = WiFi connected
- R = RFID reader OK
- ! = RFID issue

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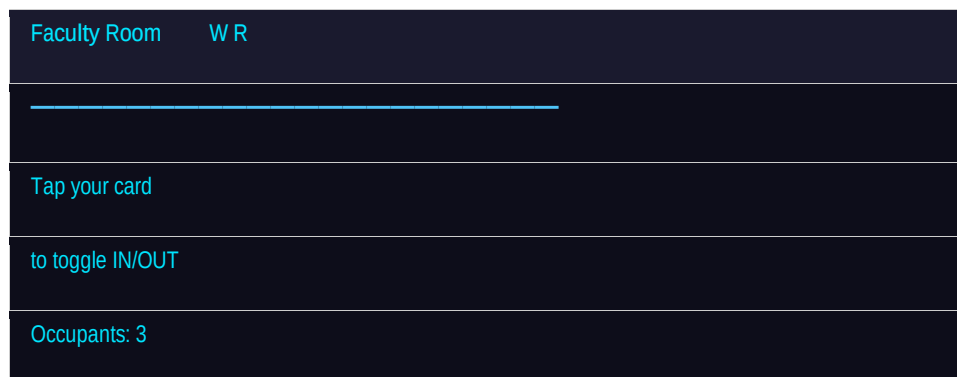
## Mode 2: Faculty Room

Purpose: Track professors entering and exiting the faculty room.

## How It Works

- Professors tap RFID card on receiver
- System toggles IN/OUT status automatically
- Logs to Firebase: /monitoring\_logs/
- Shows current occupancy count

## OLED Display



## Usage

- Professor taps card → Status changes to IN
- Professor taps again → Status changes to OUT
- Multiple professors can be IN simultaneously
- Dashboard shows who is currently in the faculty room

## Example Sequence

- Dr. Smith taps → IN (occupants: 1)
- Dr. Jones taps → IN (occupants: 2)
- Dr. Smith taps → OUT (occupants: 1)

---

## Mode 3: Registration

Purpose: Register new RFID cards.

## How It Works

- Any scanned card is sent to Firebase
- Bypasses whitelist check
- Sends data to /pending\_registrations/
- Admin approves via dashboard

## OLED Display

ADMIN MODE    W R

\_\_\_\_\_

REGISTRATION MODE

Scan card to

register

Time: 59m

## Registration Workflow

5. Admin enables registration mode
6. Professor taps their card
7. UID sent to Firebase /pending\_registrations/
8. OLED shows the UID
9. Admin sees pending registration on dashboard
10. Admin adds name and department
11. Card added to whitelist

*Note: Session timeout is 1 hour. Registration mode exits automatically after 60 minutes.*

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## Switching Modes

### Method 1: Physical Button (PROG)

Hold the PROG button for 2 seconds to cycle through modes:

- Hold PROG button for 2 seconds
- OLED shows "MODE CHANGED"
- New mode name is displayed
- Release button

Cycle order:

LoRa Monitoring → Faculty Room → Registration → LoRa Monitoring (repeats)

### Method 2: Admin RFID Card

- Tap the admin card to toggle Registration Mode ON/OFF
- Works from any mode
- Quick access for admins

Example:

- Currently in LoRa Monitoring → Tap admin card → Switches to Registration
- Tap admin card again → Returns to LoRa Monitoring

### Method 3: Dashboard Control

12. Open the web dashboard
13. Go to the Registration page
14. Click "Enable Registration Mode"
15. Receiver enters registration mode within 5 seconds
16. Click "Disable Registration Mode" when done

Firebase configuration path:

```
/system_config/registration_mode
```

- true = Registration mode ON
- false = Registration mode OFF

### Mode Comparison

| Feature           | LoRa Monitoring   | Faculty Room      | Registration            |
|-------------------|-------------------|-------------------|-------------------------|
| LoRa Reception    | Active            | Active            | Active                  |
| Local RFID        | Ignored           | Toggle IN/OUT     | Register card           |
| Whitelist Check   | N/A               | Required          | Bypassed                |
| Firebase Path     | /monitoring_logs/ | /monitoring_logs/ | /pending_registrations/ |
| Multi-User        | N/A               | Yes               | N/A                     |
| Button Switch     | Yes               | Yes               | Yes                     |
| Admin Card        | No                | No                | Toggle                  |
| Dashboard Control | No                | No                | Yes                     |

## When to Use Each Mode

### Use LoRa Monitoring when:

- Normal operation
- Monitoring individual rooms
- Receiving data from transmitters
- No local RFID needed

### Use Faculty Room when:

- Tracking faculty room occupancy
- Multiple professors sharing one room
- Need to know who is currently inside
- Professors tap on receiver directly

### Use Registration when:

- Adding new professors
- Registering new RFID cards
- Testing unknown cards
- Admin is present to approve

## GantChart

| Task/Phase                                  | Week1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---------------------------------------------|-------|---|---|---|---|---|---|---|
| Project Planning & Proposal                 | ■     | ■ |   |   |   |   |   |   |
| System Design<br>(Architecture & Data Flow) |       | ■ | ■ |   |   |   |   |   |
| Hardware Procurement<br>(RFID,ESP32,LoRa)   |       |   | ■ |   |   |   |   |   |
| RFID & ESP32 Setup                          |       |   | ■ | ■ |   |   |   |   |
| LoRa Communication Setup                    |       |   |   | ■ | ■ |   |   |   |
| Firmware Development<br>(ESP32)             |       |   |   | ■ | ■ |   |   |   |
| Database & Web Server Setup                 |       |   |   |   |   |   |   |   |
| Dashboard Development<br>(Monitoring UI)    |       |   |   |   |   |   |   |   |
| System Integration                          |       |   |   |   |   |   |   |   |
| Testing & Debugging                         |       |   |   |   |   |   |   |   |
| Documentation & Final Report                |       |   |   |   |   |   |   |   |

## Expected Output

The expected output of the IoT-Based Smart Faculty and Classroom Monitoring System using RFID Technology and LoRaWAN are the following:



### 1. Real-Time Faculty Location Display

A monitoring dashboard that shows the current location of each faculty member based on the latest RFID tap. The system will display the assigned classroom or last detected room of the faculty.

### 2. RFID-Based Identification Output

Each time a faculty member taps their RFID ID card, the system will successfully capture and display the corresponding faculty name, ID, and location.

### 3. Wireless Data Transmission Output (LoRaWAN)

The system will transmit data from classroom nodes to the faculty monitoring station using LoRaWAN, ensuring reliable long-range communication across the campus.

### 4. Monitoring Dashboard Interface

A user-friendly interface accessible to authorized personnel where they can view:

- Faculty Name
- Current Room Location
- Timestamp of last tap
- System status (Online/Offline)

### 5. Data Logging and Storage Output

All RFID tap events will be stored in a database, including:

- Faculty ID
- Room Location
- Date and Time of tap

This allows tracking of movement history for monitoring purposes.

### 6. System Status Indicators

The system will display operational status such as:

- RFID Reader Status

- LoRa Connection Status
- System Power Status

## 7. Improved Coordination Output

The system will help reduce time spent locating faculty members and improve coordination within the school by providing accurate and updated location data.

## Development Period

The development of the IoT-Based Smart Faculty and Classroom Monitoring System using RFID Technology and LoRaWAN was conducted over a period of eight (8) weeks. The process followed a structured approach, beginning from planning up to system completion and documentation.

During the first and second weeks, the researchers focused on project planning, requirement analysis, and system design. This includes the creation of system architecture, data flow, and identification of hardware and software components needed for the project.

In the third week, the group proceeded with hardware procurement and initial setup of components such as the ESP32 microcontroller, RFID (RC522) module, and LoRa module. Basic testing was also performed to ensure that each component was functioning properly.

The fourth to fifth week, the development phase was carried out. This includes firmware programming of the ESP32, integration of RFID functionality, and establishment of LoRa communication between classroom nodes and the faculty monitoring station.

The fifth to sixth week, the researchers developed the web server and monitoring dashboard. The system was configured to store and display real-time faculty location data using a database.

## **Recommended Development of IoT-Based Smart Faculty and Classroom Monitoring**

To further improve the performance, functionality, and usability of the IoT-Based Smart Faculty and Classroom Monitoring System using RFID Technology and LoRaWAN, the following development enhancements are recommended:

### **Mobile Application Integration**

Development of a dedicated mobile application is recommended to allow authorized personnel to access the monitoring system anytime and anywhere. This will improve accessibility and make real-time monitoring more convenient.

### **Real-Time Notification System**

The system may be enhanced by adding notification features such as SMS, email, or mobile alerts. This will notify administrators when a faculty member is not present in their assigned classroom or when unusual activity is detected.

### **Expansion of System Coverage**

It is recommended to expand the system by installing additional RFID readers and LoRa nodes in more classrooms and buildings. This will allow full campus monitoring and improve system effectiveness.

### **Cloud-Based Data Storage**

Integrating cloud storage is recommended to ensure data security, backup, and remote accessibility. This will also minimize the risk of data loss in case of local server failure.

### **Enhanced User Interface**

Improvement of the monitoring dashboard interface is suggested to make it more userfriendly and interactive. Adding visual elements such as charts and real-time indicators can enhance user experience.

## **Advanced Security Features**

To improve system security, additional authentication methods such as biometric verification (fingerprint or facial recognition) may be integrated alongside RFID technology.

## **Power Backup System**

The implementation of a power backup system such as UPS is recommended to ensure continuous operation during power interruptions.

## **Improved LoRa Signal Optimization**

Enhancing the placement of LoRa modules and using high-gain antennas is recommended to improve signal strength and reduce communication issues across the campus.

## **Proof of System Functionality**

The following outputs demonstrate that the IoT-Based Smart Faculty and Classroom Monitoring System using RFID Technology and LoRaWAN is functional and operating as intended:

### **RFID Tap Detection Proof**

Screenshots or photos showing the successful detection of RFID cards when tapped on the RC522 reader, including the captured UID displayed on the system.

### **Real-Time Location Update Proof**

Images of the monitoring dashboard showing the immediate update of faculty location after an RFID tap, including room assignment and timestamp.

### **LoRa Communication Proof**

Photos or logs demonstrating successful wireless data transmission from classroom nodes to the faculty monitoring station using LoRa modules.

### **ESP32 Data Processing Proof**

Captured serial monitor output showing the processed data (faculty ID, room, timestamp) before transmission.

### **Dashboard Monitoring Interface Proof**

Screenshots of the web-based dashboard displaying faculty name, current location, last detected room, and system status in real-time.

### **Database Logging Proof**

Screenshots of the database (e.g., MySQL/Firebase) showing stored records of RFID tap events including date, time, and location.

### **System Status Indicators Proof**

Images showing system status such as LoRa connectivity, system ON/OFF state, and successful tap indicators.

### **Multiple Classroom Node Functionality Proof**

Photos showing multiple RFID nodes installed in different classrooms, demonstrating simultaneous operation and data transmission.

### **End-to-End System Operation Proof**

A complete sequence of images showing the full process: RFID tap → data processing → LoRa transmission → dashboard update.